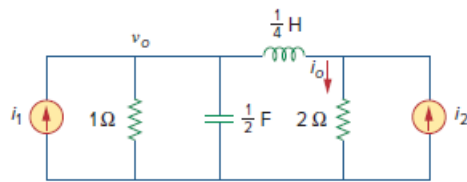


Problem

For the electric circuit in Fig. 16.25, determine the state model. Take v_o and i_o as the output variables.

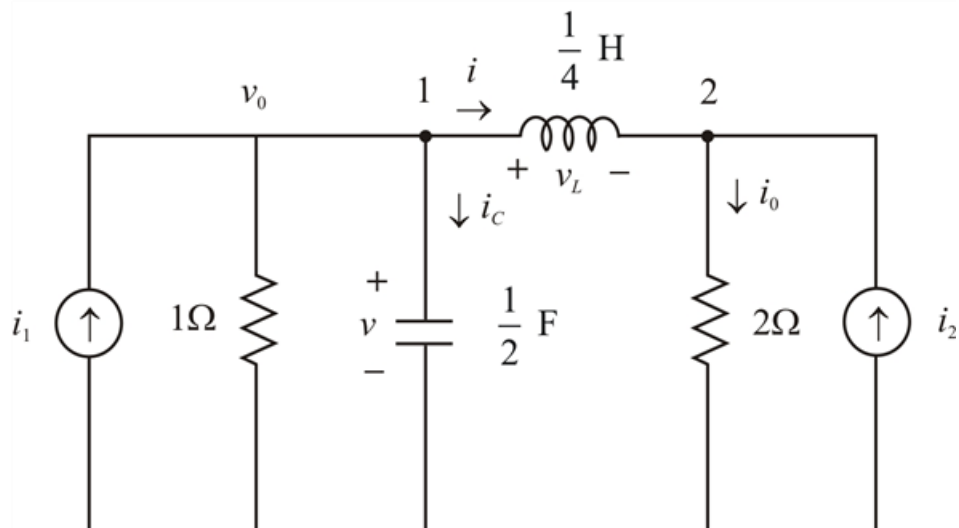
Figure 16.25:



Step-by-step solution

Step 1 of 4

The given circuit



Step 2 of 4

We know the

$$v_L = L \frac{di}{dt}$$

$$i_C = C \frac{dv}{dt}$$

Here two inputs are i_1 and i_2 and two outputs are v_0 and i_0 . We select the inductor current i and capacitor voltage v as the state variables.

Applying KCL at node 1

$$-i_1 + \frac{v_0}{1} + i_C + i = 0$$

$$i_C = i_1 - i - v_0$$

We know that

$$v_0 = v \quad \text{---(1)}$$

$$C \frac{dv}{dt} = i_1 - i - v$$

$$\frac{dv}{dt} = \frac{i_1}{C} - \frac{i}{C} - \frac{v}{C}$$

Step 3 of 4

By substituting the values of $C = \frac{1}{2}$, we get

$$\dot{v} = \frac{i_1}{\frac{1}{2}} - \frac{i}{\frac{1}{2}} - \frac{v}{\frac{1}{2}}$$

$$\dot{v} = 2i_1 - 2i - 2v \quad \text{---(2)}$$

Applying KCL at node 2, we get

$$i_0 - i_2 - i = 0$$

$$i_0 = i_2 + i \quad \text{---(3)}$$

Step 4 of 4

Applying KVL around the loop containing $\frac{1}{2}\text{F}$ capacitor, $\frac{1}{2}\text{H}$ inductor and 2Ω resistor.

$$-v + v_L + Ri_0 = 0$$

$$v_L = v - R(i_2 + i) \quad [\because \text{from (3)}]$$

$$L \frac{di}{dt} = v - Ri_2 - Ri$$

$$\frac{di}{dt} = \frac{v}{L} - \frac{R}{L}i_2 - \frac{R}{L}i$$

$$L = \frac{1}{4} \text{ H}, R = 2\Omega$$

By substituting the values, we get

$$\dot{i} = \frac{v}{\frac{1}{4}} - \frac{2}{\frac{1}{4}}i_2 - \frac{2i}{\frac{1}{4}}$$

$$\dot{i} = 4v - 8i_2 - 8i \quad \text{--- (4)}$$

Putting equations (1), (2), (3) and (4) together in the standard form leads to the state model for the circuits.

$$\begin{bmatrix} \dot{v} \\ \dot{i} \end{bmatrix} = \begin{bmatrix} -2 & -2 \\ 4 & -8 \end{bmatrix} \begin{bmatrix} v \\ i \end{bmatrix} + \begin{bmatrix} 2 & 0 \\ 0 & -8 \end{bmatrix} \begin{bmatrix} i_1 \\ i_2 \end{bmatrix}$$

$$\begin{bmatrix} v_0 \\ i_0 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} v \\ i \end{bmatrix} + \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} i_1 \\ i_2 \end{bmatrix}$$